Networking Lab Cheat Sheet

<u>IP Section:</u>

Command	Explanation
ip a	Displays all network interfaces with their **IP addresses, MAC addresses, and status**.
ip a show dev eno2	Shows detailed IP configuration **only for the network interface eno2**.
sudo ip a add 172.17.100.19/24 dev eno2	Assigns the IP address **172.17.100.19** with subnet mask **/24** to interface **eno2**.
sudo ip a del 172.17.100.19/24 dev eno2	Removes the assigned IP address from interface **eno2**.
ping 10.34.0.142	Sends continuous ICMP packets to test connectivity to the destination.
ping –c 5 10.34.0.142	Sends **only 5 ICMP packets** then stops.
ping –n –c 5 10.34.0.142	Sends 5 packets and **does not resolve hostnames** (shows numeric IP only).
ping –n –c 5 –R 10.34.0.142	Sends 5 packets, does not resolve names, and **records router path (route tracing)** if supported.
ping –I 172.17.100.19 10.34.0.142	ping –I <source> <destination>
ip r	Displays the **current routing table**
sudo ip r add 172.17.100.19/32 via 10.34.0.143 dev eno2	Adds a **route to a specific host (172.17.100.19)** through gateway **10.34.0.143** using network interface **eno2**.
sudo ip r del 172.17.100.19/32	Deletes the **specific routing rule** for that host from the routing table.
ip neigh	Shows the **ARP table** (maps IP addresses to MAC addresses of nearby devices).

ipcalc 172.17.100.19/24	Shows ip calculation of this host and some other stuff

<u>SSH Server:</u>

Command Sequence	Purpose
whoami	Shows username
sudo apt install openssh-server	Does what it says
which sshd	Shows the location of the sshd executable file in the system path.
sudo systemctl start sshd	Starts the ssh server for remote connection
sudo systemctl status sshd	Shows server status
ssh username@localhost	Connects to the local machine's **SSH server** for secure remote login testing.
cd /etc/ssh	Go to the ssh config directory
sudo nano sshd_config	Uncomment the port, and change it e.g Port 2233
sudo systemctl restart sshd	Restarts ssh service
ssh –oPort=2233 username@ip	Connect to SSH using a **custom port**
ssh –p 2233 username@ip	Does the same thing
sudo reboot	📋 📋 📋
logout	Closes connection
sudo apt purge openssh-server	

Bind:

Command Sequence	Purpose
sudo apt install bind9	Duhhh
which named	Shows the **location of the DNS server daemon (named) executable** in the system.
sudo systemctl start bind9	Start DNS Service (Ubuntu)
sudo systemctl status bind9	Check Status
cd /etc/bind/	Go to bind directory
sudo cp db.local db.s19.zhjlab.bd	Copy
sudo nano db.s19.zhjlab.bd	Change stuff here first

Change it to

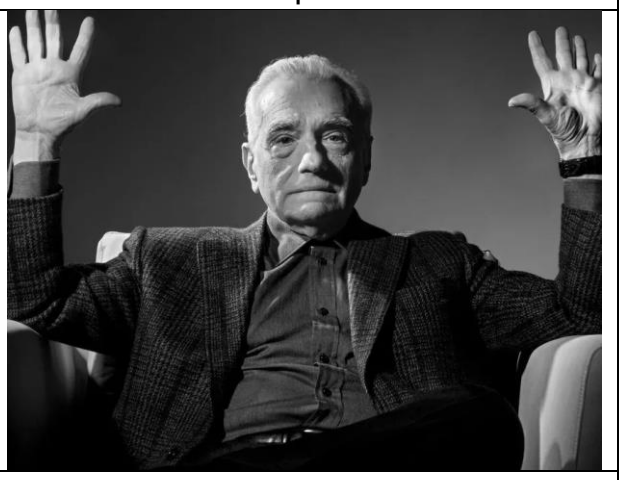
```
$TTL        300
@           IN    SOA   ns1.s19.zhjlab.bd.   bsse1619.iit.du.ac.bd.  (
                                                   3  ; Serial
                                              604800 ; Refresh
                                               86400 ; Retry
                                             2419200 ; Expire
                                              604800 ) ; Negative Cache TTL
;
@                       IN    NS        ns1.s19.zhjlab.bd.
ns1.s19.zhjlab.bd.      IN    A         172.17.100.19
www                     IN    CNAME     ns1
```

sudo nano /etc/bind/named.conf.local	Add here

```
zone "s19.zhjlab.bd" IN {
   type master;
   file "/etc/bind/db.s19.zhjlab.bd";
   allow-update { none; };
};
```

sudo named-checkzone s19.zhjlab.bd /etc/bind/db.s19.zhjlab.bd	Check zone file If “Ok” proceed
sudo tail -f /var/log/syslog	Run this in another terminal tab and monitor this
sudo systemctl restart bind9	Restart bind service
dig –t soa s19.zhjlab.bd@172.17.100.19	Check

<u>Apache:</u>

Command Sequence	Purpose
sudo apt install apache2	
sudo systemctl start apache2	Starts the apache service

localhost	Type in browser (verification)
cd /var/www/html	Go to document root
sudo nano index.html	Write an html file

```
<html>

<head>

<title>OOOOOOOOOOOOOOOOOOOOOOO</title>

</head>

<body>

<h1>UUUUUUUUUUUUUUUUU</h1>

<p>I wanna die</p>

</body>

</html>
```

sudo systemctl restart apache2	Restarts the service
sudo mkdir /var/www/mysite	Creates mysite directory
sudo cp /var/www/html/index.html /var/www/mysite/index.html	Copies the index html file to mysite directory
sudo cp /etc/apache2/sites-available/000-default.conf /etc/apache2/sites-available/001-mysite1.conf	Copy the config file
sudo nano /etc/apache2/sites-available/001-mysite1.conf	Edit

Change ServerName part to

ServerName s19.zhjlab.bd

and

Change document root from

/var/www/html

to

/var/www/mysite

sudo nano /var/www/mysite/index.html	Change some stuff here
sudo a2ensite 001-mysite1.conf	Enable the site
sudo a2dissite 000-default.conf	Better to disable the default site(optional)
sudo nano /etc/hosts	Edit

Add
172.17.100.19 s19.zhjlab.bd

sudo systemctl restart apache2	Restarts the service
http://s19.zhjlab.bd	For verification